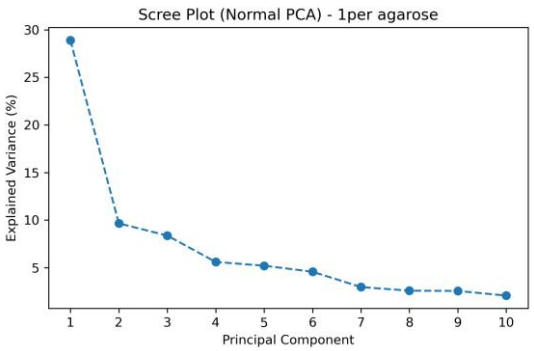
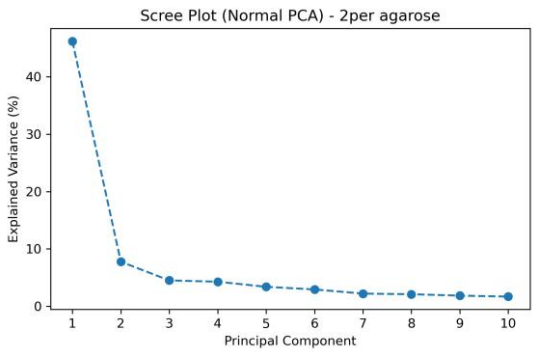
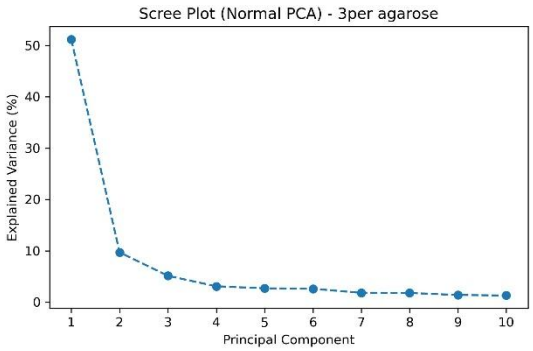
First, PCA was done on the dataset so that neurons are the features. I also checked out of curiosity the scree plots:



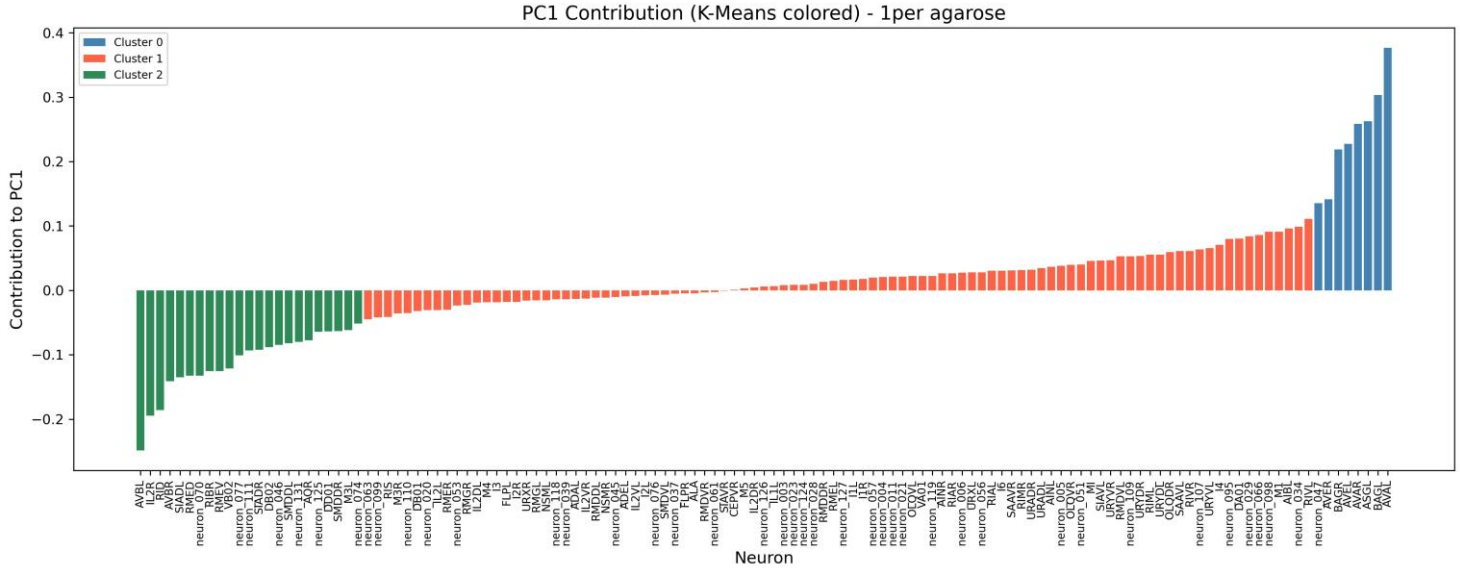


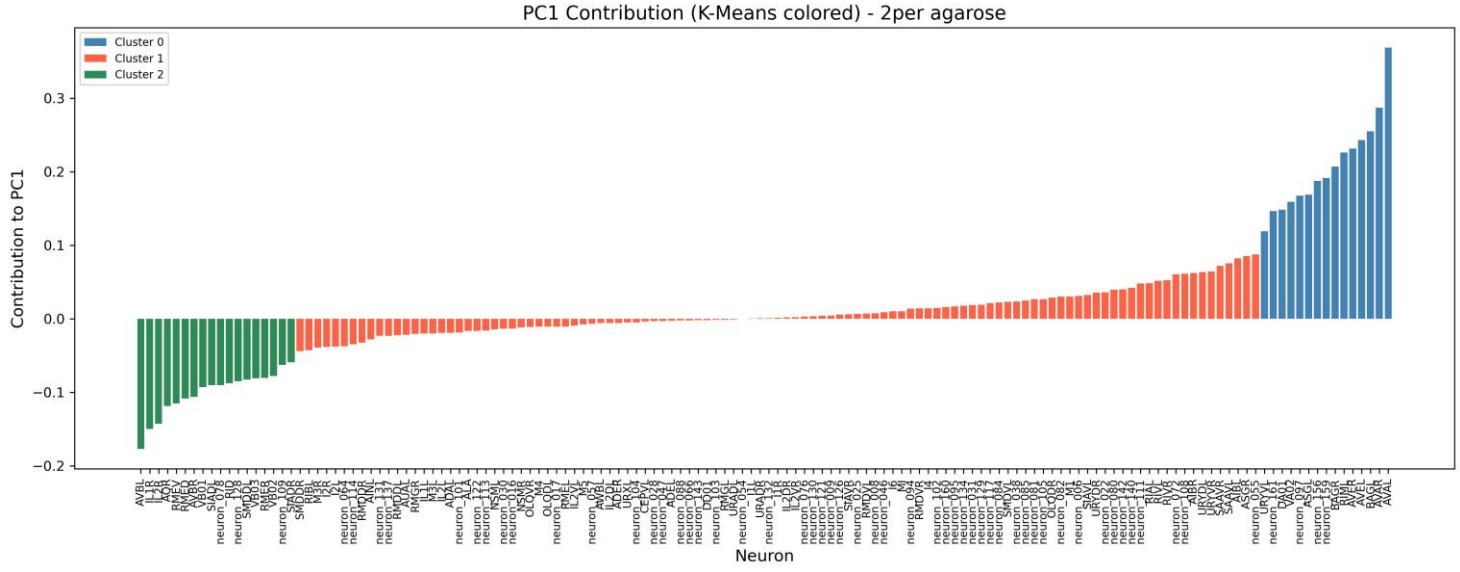


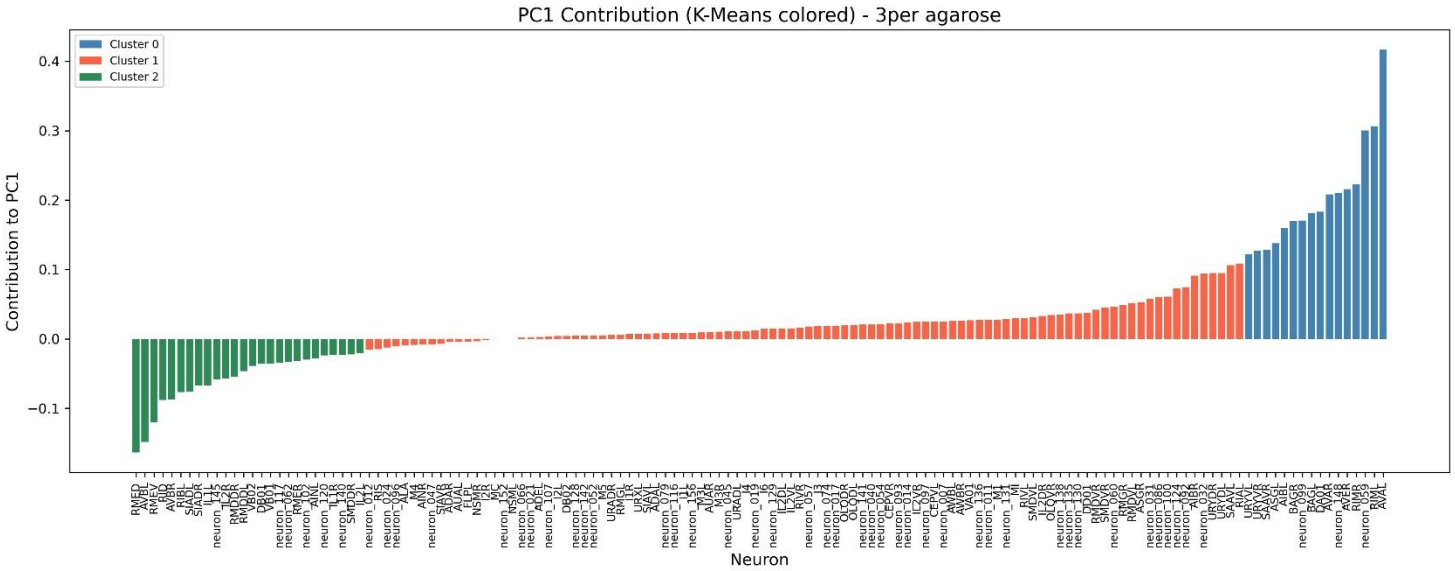
As we can see in 1per the first pc explains the least of the variance compared to the other conditions and intuitively based on the graph - 2 or 3 seem like the best numbers.

PC1

Here are the loadings of the neurons for PC1, colored by k means of 3:



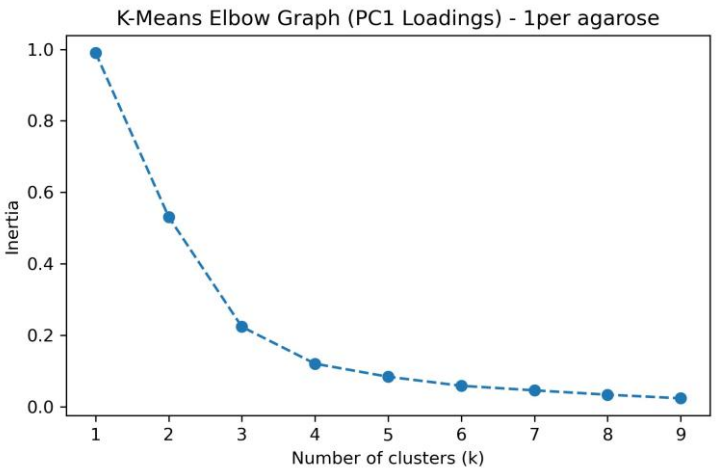


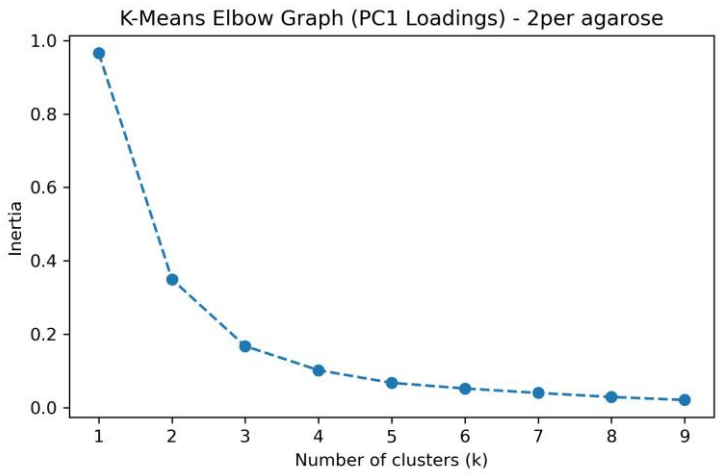


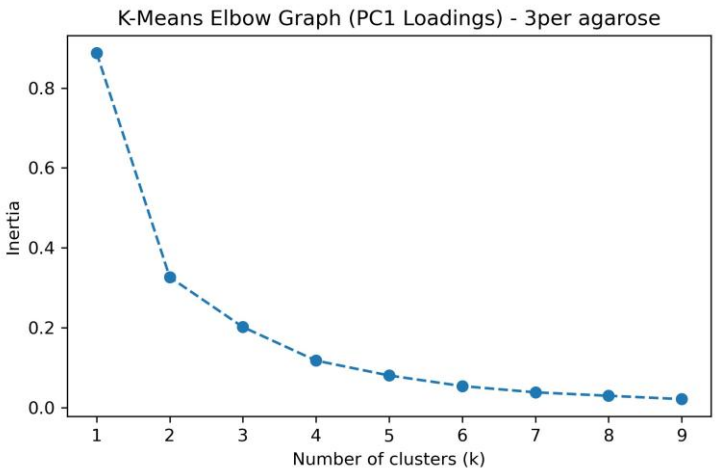
Value of 3 for k was chosen both intuitively after our zoom call where we discussed the 3 types of neurons:

- Reverse
- Forward
- Unrecognized

and also by and by ths elbow graph:







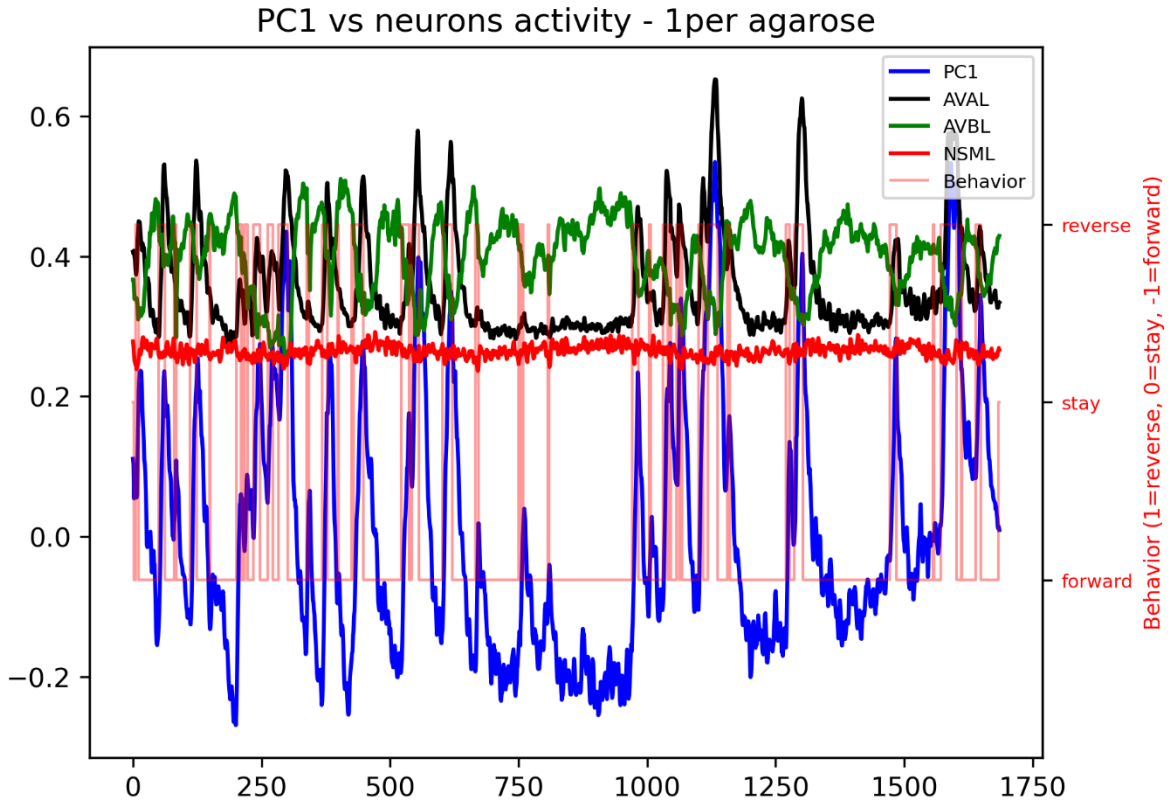
We can tell that **generally**:

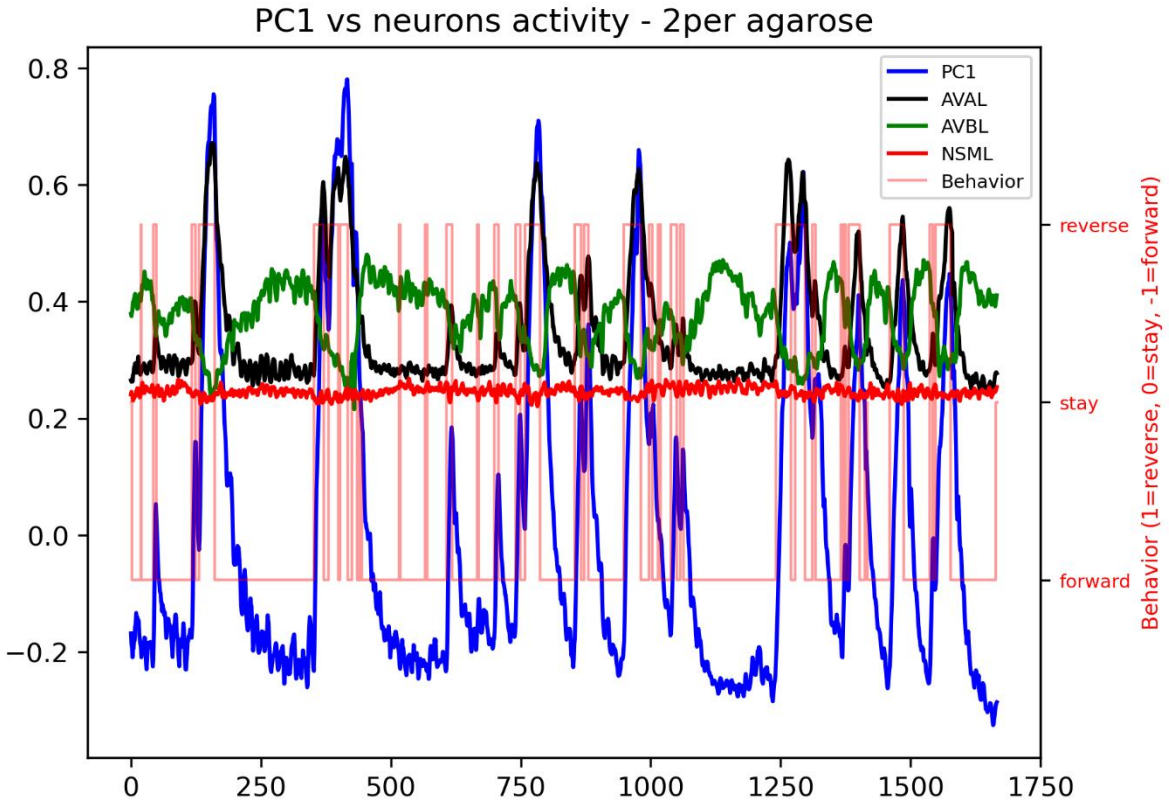
- The blue group are the reverse neurons
- The green group are the forward neurons
- The orange group are neither (because of the rough k means clustering its fair to say that maybe some of them are actually reverse/forward.

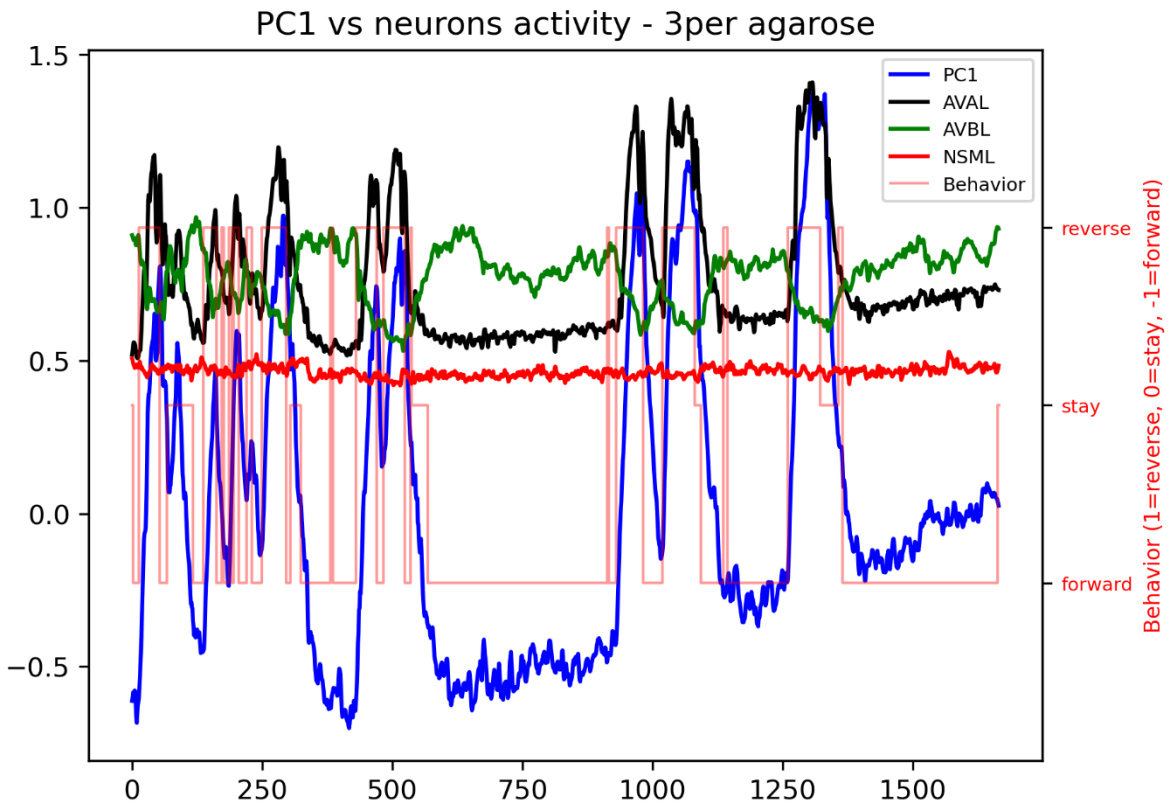
The number of neurons that didn't switch sides - meaning were green/blue in all of conditions are:

- always blue(reverse) cluster - 13
- always green(forward) cluster - 22

I think positive contributions to pc1 represent reverse neurons and negative represent forward neurons. I arrive at this conclusion by looking at the activities of a few neurons from each group – some are shown in the following graph that shows the activities of: AVB which is constantly (strong negative) green, AVA which is constantly (strong positive) blue, NSML which has close to zero/exactly zero contribution, together with PC1 and with the behavior itself.

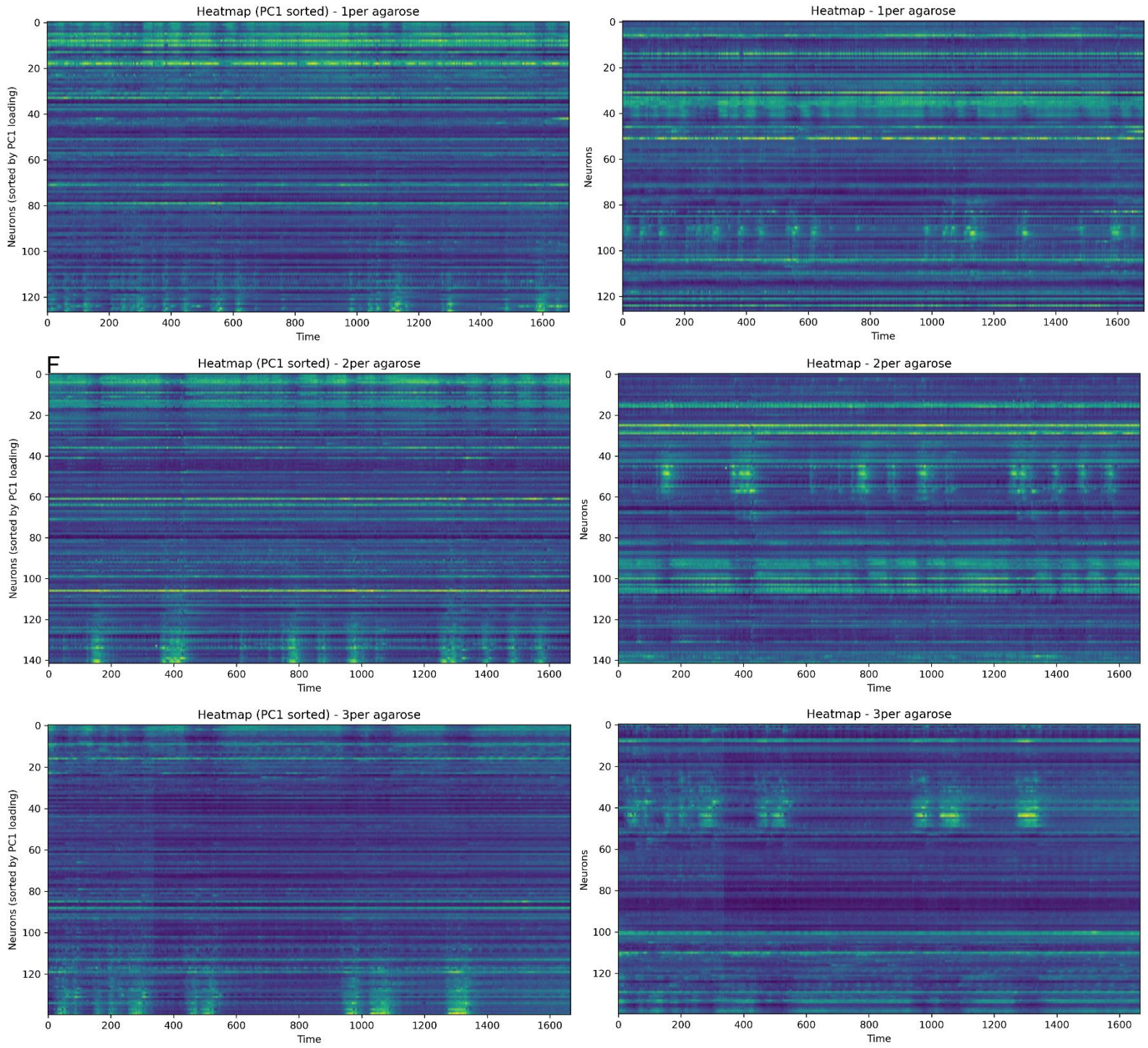




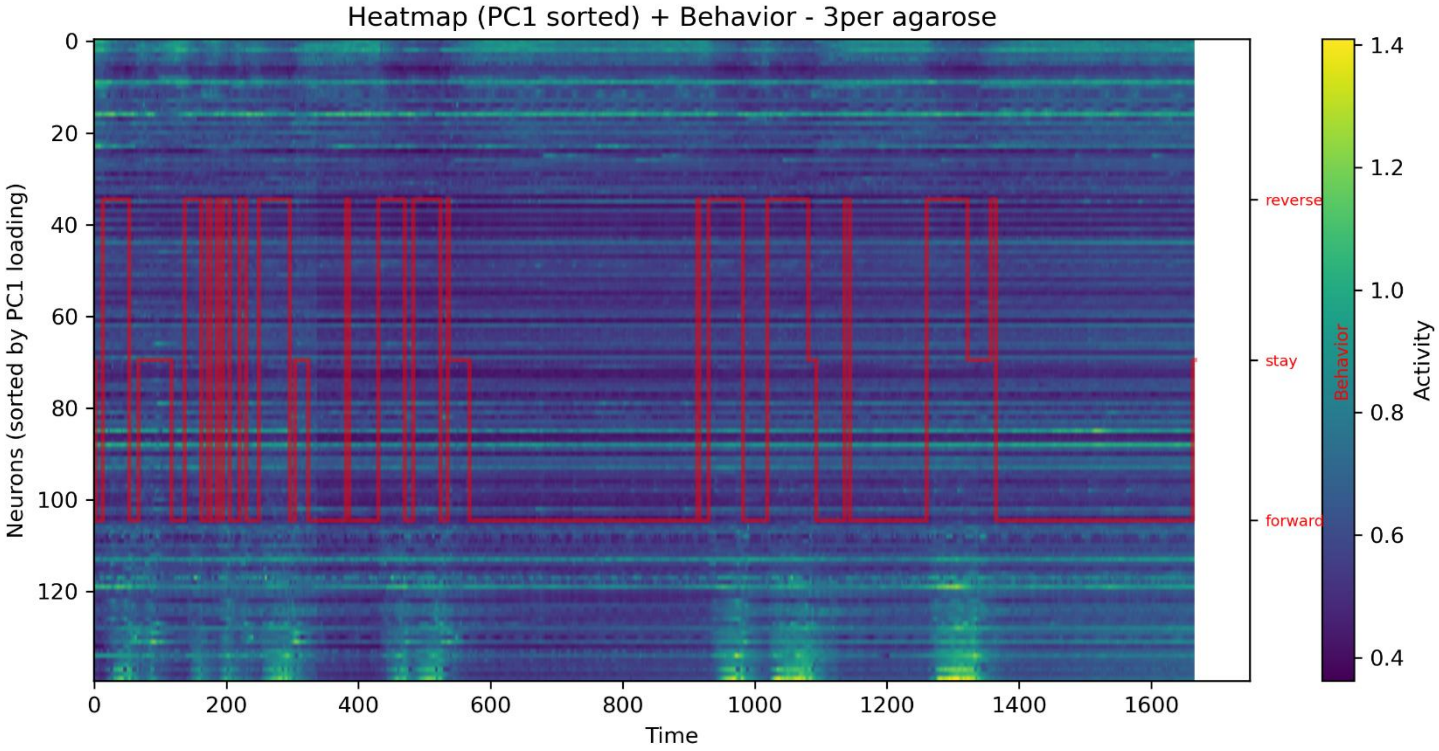


We can see that AVAL has a strong correlation to PC1 while AVBL has a negative strong correlation and NSML, at least from this graph, looks a bit like a dead weight for the worm.

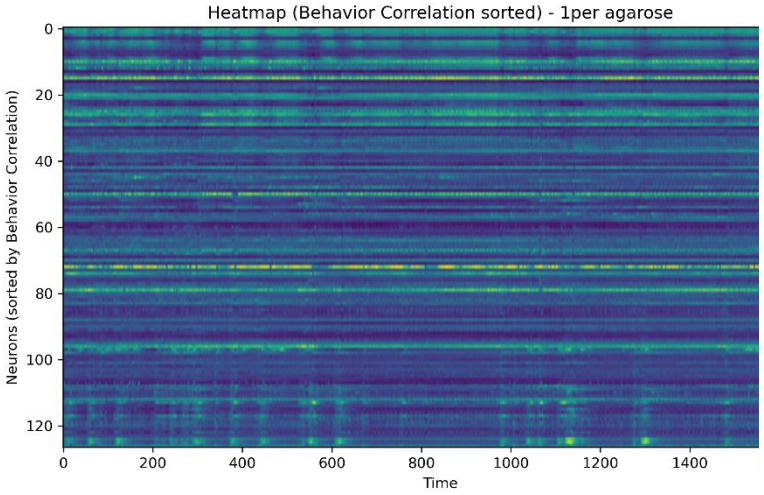
Finally, here shown a heatmap of the neurons activities ordered by the pairwise pearson correlation (right) side by side with another heatmap ordered according to the loadings of PC1 (left):

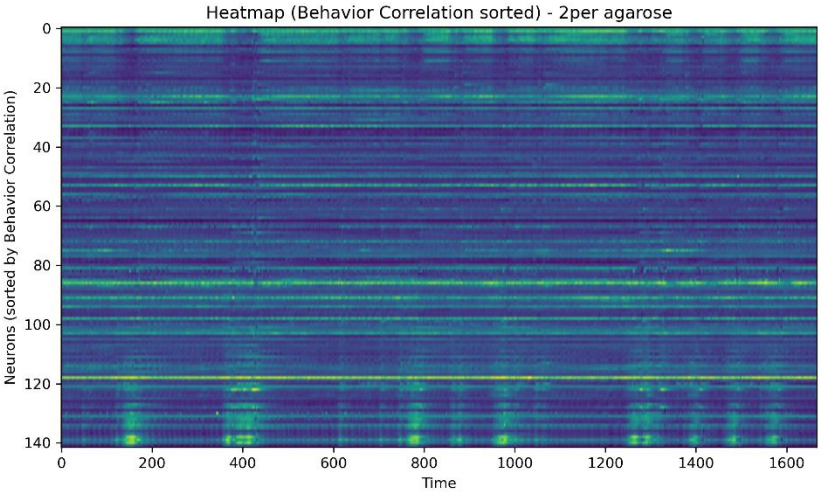


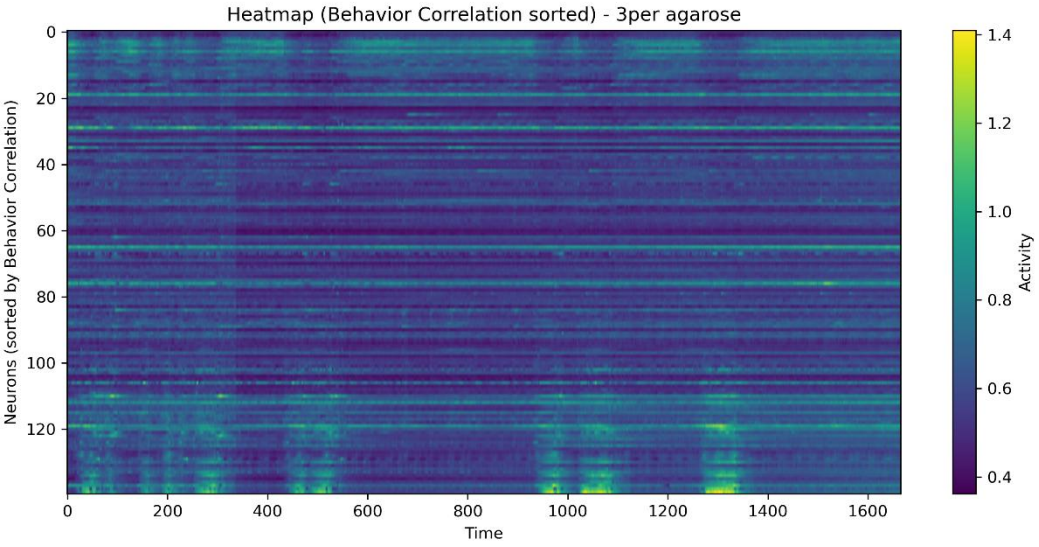
One small thing off the bat is that condition 3 looks slightly “darker” which may mean less neuronal activity or that something biological or measurement related led to lower ratio overall.
It makes sense but still surprised me a bit to see how “smooth” looking are the PC1 ordered heatmaps. It highlights (no pun intended) how PC1 captures the connection between the neurons that fire together for meaningful periods, and also the connection between these who don’t - It’s easy to see two main groups of neurons (in the PC1 ordered they are in the top and bottom) that fire “alternately”.
Something I have found really cool is plotting the behavior on top of the heatmap. Here is an example with condition 3 where you can see it the clearest:



Next, I created yet another heatmap but now ordered by the pearson correlations of the neurons not with themselves but with the behavior, which can be thought of as a graph with 3 values (reverse, forward and staying at the same place).



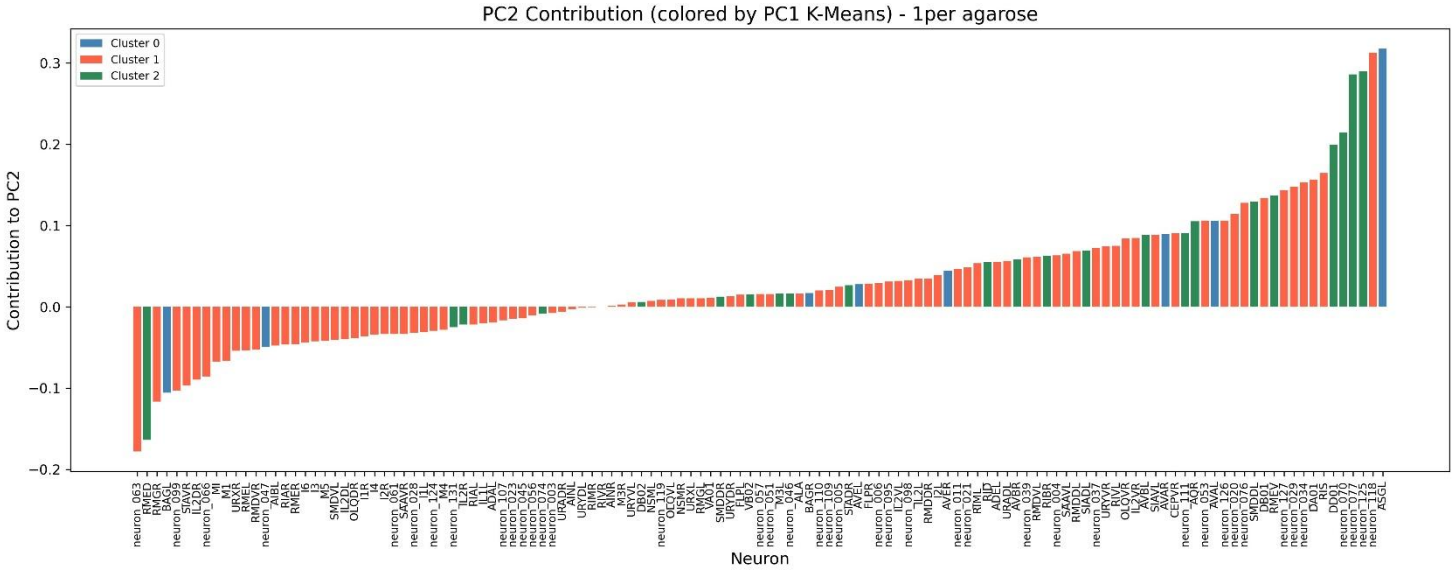


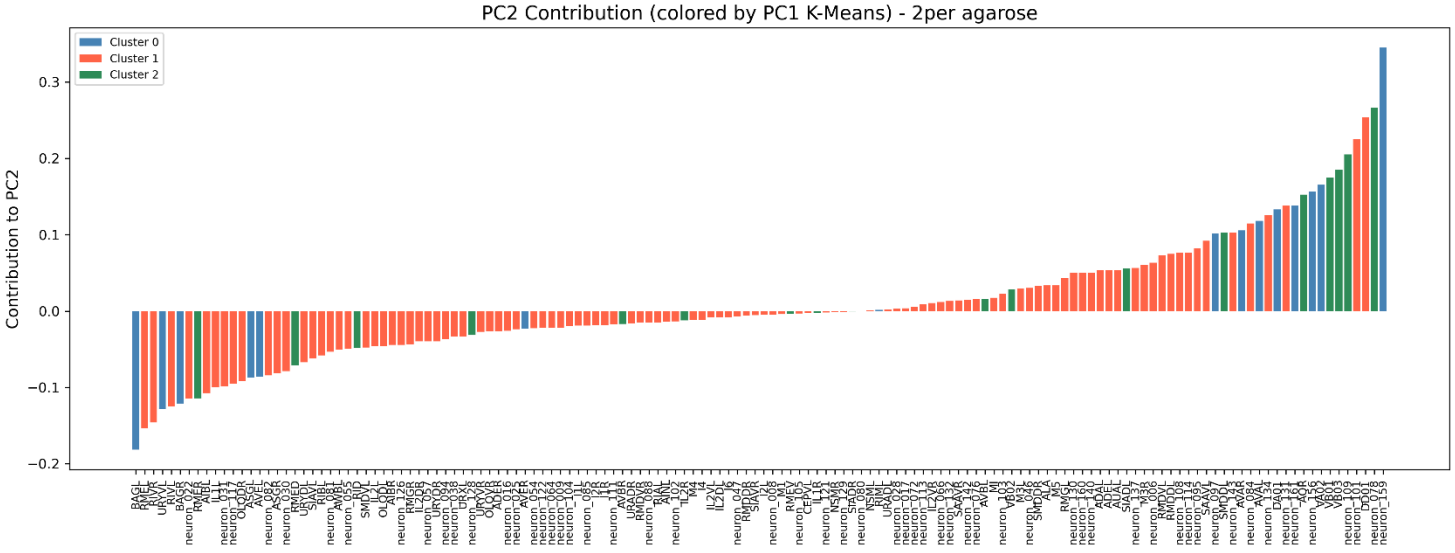


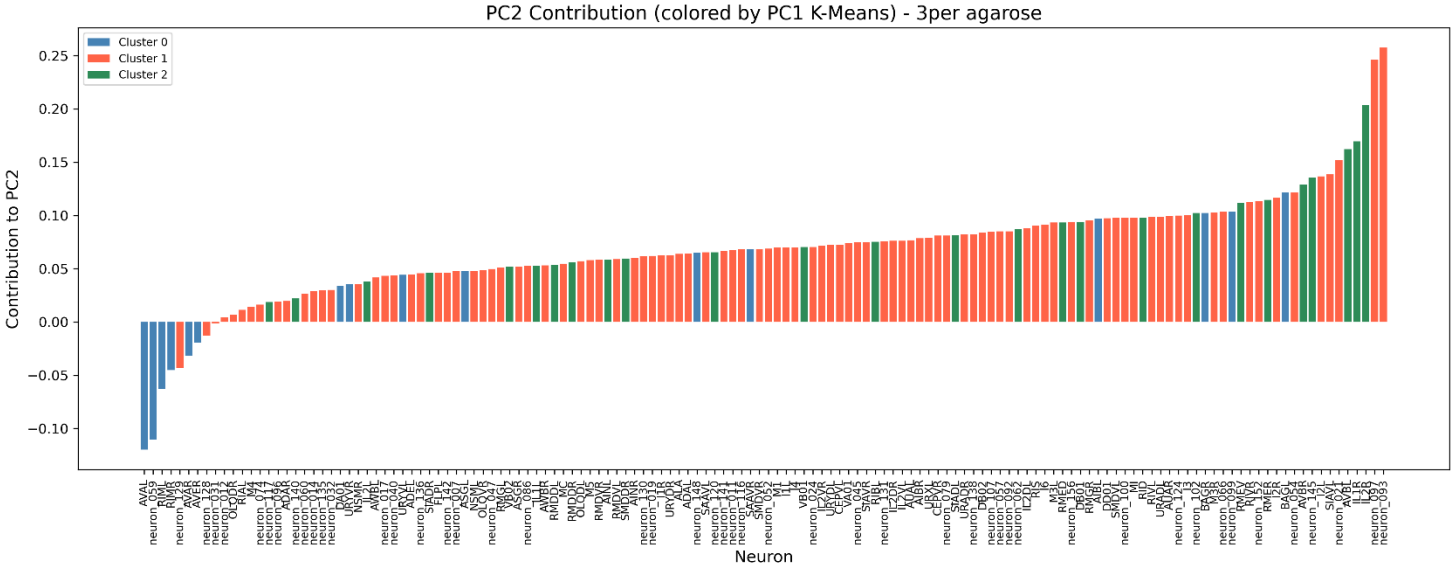
I guess it can be used to group neurons too but unlike PCA it presupposes them to the reverse/forward behavior. It still looks similar to the PC1 ordered heatmaps which support the idea of PC1 representing reverse/forward.

PC2

Next I tried to understand what PC2 represents but I was unsuccessful.
Here are the contributions of the neurons to PC2, colored according to the groups found earlier with k-means on PC1.

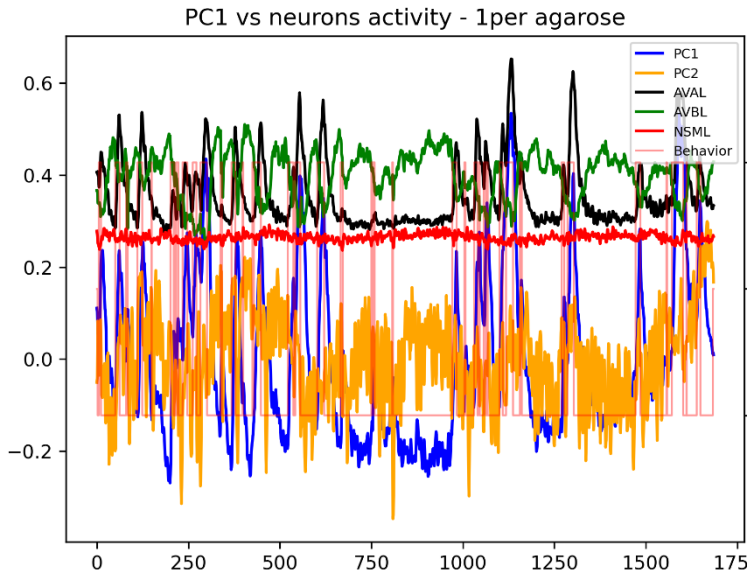


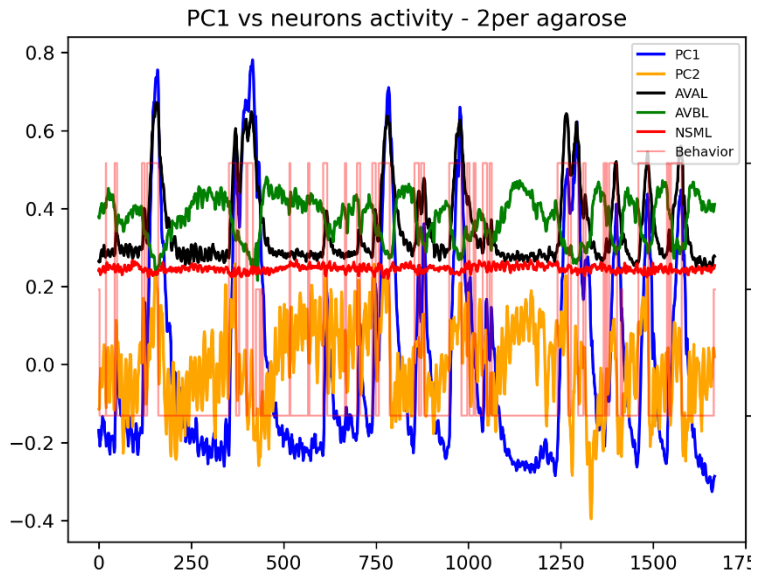


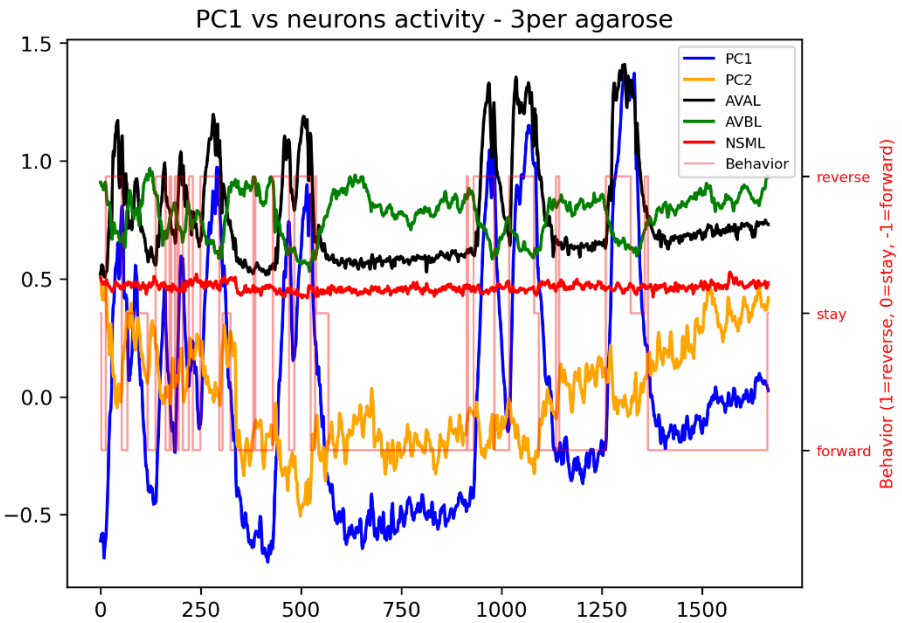


First of all, the “shapes” themselves of the graphs are not similar to each other at all (especially condition 3), unlike with PC1 where the graphs look alike.
Second, the supposed reverse and forward neurons seem to be scattered randomly here (which when I think about it makes sense duo to how PCA works).

The activity of PC2 compared to PC1 also doesn’t give anything.

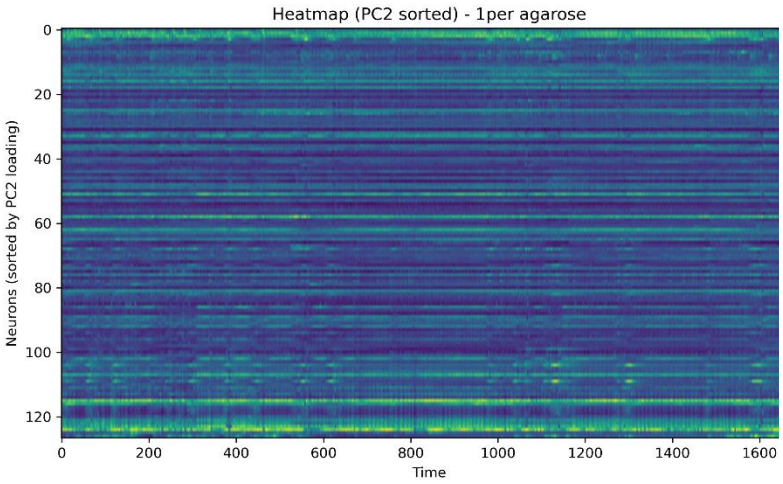


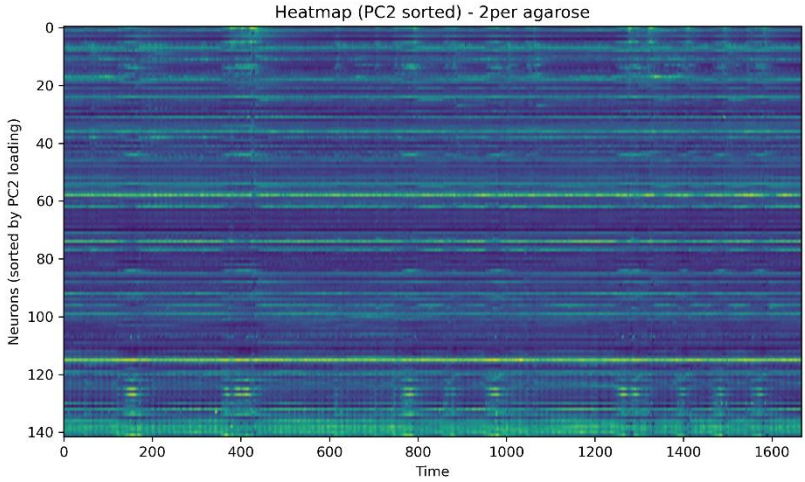


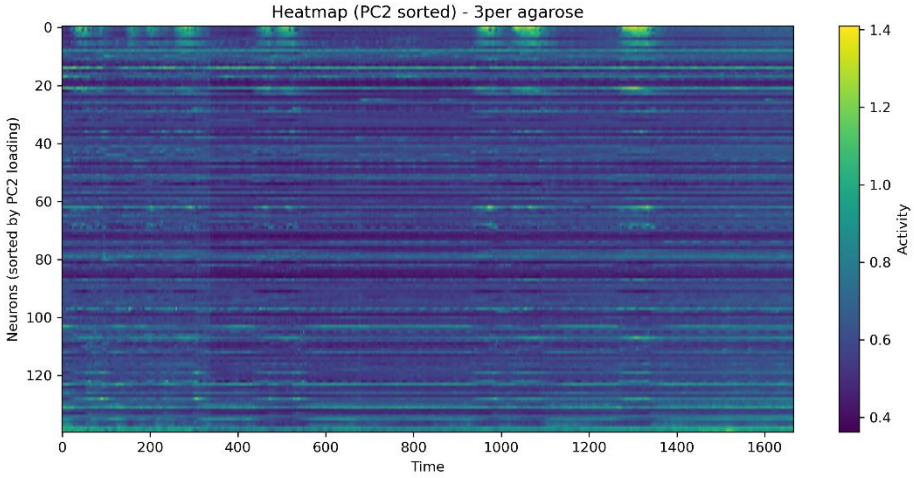


Most of the time it seems to move with AVBL (especially in condition 3) but sometimes it’s not and moves with PC1.

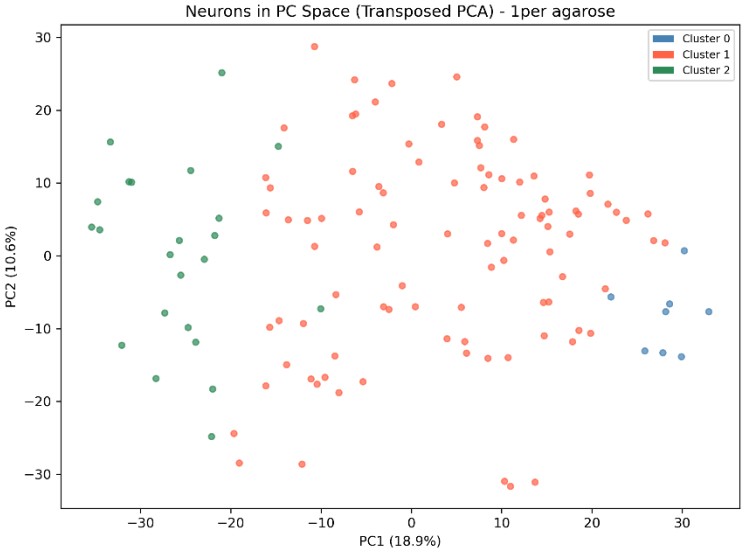
Heatmaps ordered by PC2 contributions look like the distorted version of the “pc1 loadings order” heatmaps from before.

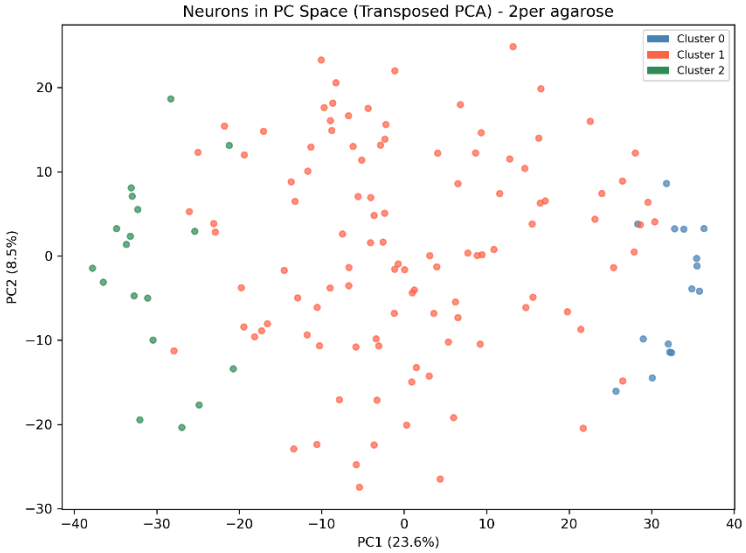


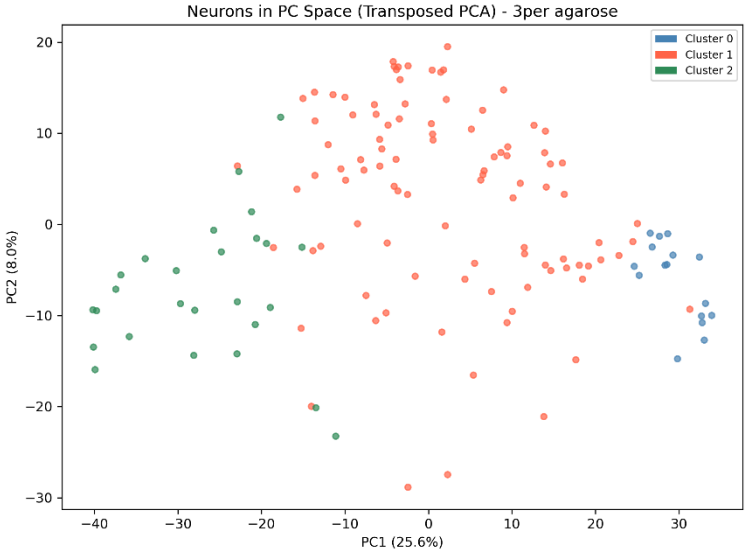




One last try was doing PCA on the transposed matrix of the traces where the features are time units and samples are the neurons. Here are scattered plots where every dot is a neuron, colored by the original groups from the k-means earlier, so blue are reverse neurons, green are forward and orange are unknown.







Expectedly, PC1 here represents the times when the worm reverses so the reverse (blue) neurons have higher ratios and the forward (green) lower. I thought I might see something interesting in PC2 and understand what kind of time/events it represents but the only thing that pops to my eyes is the fact that green dots are more spread apart than blue dots on PC2 axis.